

Balance Through Constrained Procedural Content Generation in Super Auto Pets

Final Report - Undergraduate Thesis

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Glossary

PCG: Procedural Content Generation

RL: Reinforcement Learning

SAP: Super Auto Pets

Structured Abstract

Context and motivation:

This thesis investigates whether constrained procedural generation of new Super Auto Pets-like artifacts can improve balance in a fixed, reproducible simulation environment without collapsing strategic diversity.

Research Question/Objectives:

The primary question is whether safe generated artifacts can improve simulator-backed balance outcomes while preserving diversity. Secondary questions examine whether iterative generation outperforms one-shot generation and whether selected artifacts generalize across policy families, seeds, and packs.

Principal ideas:

The approach combines deterministic frozen-environment execution, constrained generated-artifact design, deterministic artifact injection, and simulator-backed evaluation of fairness, diversity, stability, safety, and efficiency.

Research Methodology:

A frozen Super Auto Pets environment was extended with generated-artifact injection and evaluated through a final experiment matrix covering no-generation, baseline generation, generator selection, one-shot generation, iterative balancing, and transfer conditions across multiple opponent policy families.

Principal results:

The results support a cautious positive answer to the primary question. Safe generated artifacts produced modest improvement over a fixed heuristic baseline under a matched simulator-backed protocol while preserving high diversity retention. However, the gains were incremental rather than dramatic, one-shot generation outperformed the tested iterative procedure, and robustness across packs and policy families was partial rather than universal.

Principal novelty:

The main novelty is a reproducible auto-battler testbed that evaluates balance-through-generation by injecting constrained new content directly into a deterministic Super Auto Pets simulator.

Impact of results:

These findings support extending content-addition-based balancing to the auto battler setting, but only with narrow claims about effect size, safety, and generalisation.

Limitations:

The policy-family set still uses scripted_value rather than a trained learned agent, the seed count is below the original ideal target, and robustness remains artifact-dependent rather than invariant.

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1. Introduction

Balancing complex strategy games is difficult because even small changes can affect player choices, strategic diversity, and overall fairness. Previous research has shown that this problem can be studied computationally rather than only through repeated manual tweaking (Beau & Bakkes, 2016; DeLaurentis et al., 2021). In many live-service games, however, balance is still handled mainly through buffs (increasing power or effectiveness) and nerfs (decreasing power or effectiveness) to existing content. That approach can work, but it can also be time-consuming, difficult to audit, and disruptive to players. An alternative is to add carefully designed new content that changes the strategic environment without directly rewriting older content.

This thesis applies that idea to the auto battler *Super Auto Pets* (SAP). It asks whether constrained procedural generation of new SAP-like artifacts can improve simulator-backed balance outcomes while preserving strategic diversity in a fixed, reproducible environment. The project is motivated by prior work showing that game balance can be approached through content generation rather than only through parameter changes (Babin & Katchabaw, 2025). At the same time, auto battlers are a harder setting because they combine stochastic shops, tier progression, team synergies, and strong links between early-game tempo and late-game scaling, resulting in a larger number of interacting variables.

The main contribution of this thesis is not just a generator. It is a reproducible simulator-backed framework for studying balance in SAP. The project combines deterministic environment execution, constrained artifact generation, safe artifact injection, baseline comparison, and structured evaluation across policies, seeds, and packs. This matters because balance claims are much stronger when the generated content is evaluated inside the simulator rather than only through proxy scores.

Three research questions guide the work. The primary question asks whether constrained procedural generation of new SAP-like artifacts can reduce imbalance while preserving diversity. The first secondary question asks whether iterative generation outperforms one-shot generation. The second asks whether selected artifacts generalize across opponent policies and random seeds. The final results support a cautious positive answer to the primary question, do not support iterative generation outperforming one-shot generation, and support only a qualified robustness claim.

The rest of the thesis is structured as follows. Section 2 reviews relevant work on automated game balancing, procedural content generation, and the supervisor’s content-addition framing. Section 3 defines the research goal and objectives. Section 4 presents the methodology, including the frozen SAP environment, artifact schema, injection mechanism, baselines, metrics, and experiment design. Section 5 reports the empirical results. Section 6 discusses implications, limitations, and generalizability. Section 7 concludes. Section 8 outlines future work and lessons learnt. Supporting details are provided in later sections and the appendix, including detailed definitions of the key evaluation criteria and reported metrics.

2. Background and Related Work

2.1 Automated Game Balancing

Automated game balancing has often been framed as a search, optimization, or engineering-design problem. Beau and Bakkes (2016) demonstrate that balance in asymmetric games can be explored computationally rather than only by manual design iteration. DeLaurentis et al. (2021) further argue that balancing can be treated as a systematic engineering design task with explicit goals, trade-offs, and evaluation procedures. These works motivate the thesis perspective that balance should be studied through repeatable computational workflows rather than through informal tweaking alone.

At the same time, much of this prior work focuses on adjusting existing systems, parameters, or content. That leaves open an important question: can balance be improved by adding new constrained content instead of only modifying the old content base? This question is particularly relevant in complex games where changing familiar content can have disruptive or difficult-to-predict side effects.

2.2 Procedural Content Generation for Balance

Procedural content generation (PCG) research offers a natural framework for addressing balance through controlled artifact creation. PCGRL shows that content generation can be treated as an optimization problem with validity constraints and explicit objective shaping (Khalifa et al., 2020). That perspective is useful here because it suggests that generated artifacts do not need to be arbitrary or unconstrained; they can be produced within a deliberately limited design space and evaluated against measurable goals. More broadly, dynamic content-generation work such as *Galactic Arms Race* shows that generated artifacts can be adapted over time in response to gameplay, even when the original goal is personalization rather than balance (Hastings et al., 2009). This is relevant because it demonstrates that content generation can be used to shape an evolving game ecosystem rather than merely produce static assets offline.

The most important direct influence on this thesis is Babin and Katchabaw (2025), which frames game balance through procedural content generation and argues that adding new content can restore equilibrium without directly editing the existing content set. That work provides the conceptual foundation for this thesis. Their approach uses reinforcement learning (RL) in a simplified Pokémon environment, drawing on the broader idea of sequential decision making in dynamic environments (Kaelbling et al., 1996) and on deep RL methods that have been shown to scale to more complex tasks (Mnih et al., 2016). However, the current thesis is not a direct reimplement of the same method. Instead, it adapts the core idea to a different genre and a different engineering pipeline. Whereas the supervisor paper used learned embeddings and reinforcement learning in a simplified setting, this thesis uses constrained search with simulator-backed final evaluation in an auto battler.

2.3 Auto Battlers and Super Auto Pets as a Test Domain

Auto battlers are a useful stress test for balance-through-generation because they combine stochastic drafting, economy management, team composition, and nonlinear synergies. *Super Auto Pets* is especially appropriate because players build teams from a rotating shop, manage

gold under turn limits, and create strong compositions through layered interactions among pets, food, levels, and scaling effects (Team Wood Games, n.d.). A small change in available content can alter tempo, economy, and composition diversity simultaneously.

This complexity creates both opportunity and risk. On the one hand, adding the right artifact could create new counterplay or improve fairness. On the other hand, a poorly constrained artifact could create degenerate strategies or brittle interactions. For this reason, the thesis uses a frozen simulator snapshot and strict artifact constraints to keep the problem controlled and auditable.

2.4 Analysis and Research Gap

The gap addressed by this thesis is not simply whether PCG can generate game content. The more specific question is whether constrained content generation can be used as a balance intervention in an auto battler while still supporting reproducible evaluation. Existing work supports automated balancing and PCG as broad ideas, and the supervisor paper supports balancing by adding content, but there is still limited evidence on whether that framing transfers to SAP-like auto battlers under deterministic simulator-backed evaluation.

This thesis addresses that gap by building an end-to-end pipeline that generates constrained SAP-like artifacts, injects them into a frozen simulator, compares them against required baselines, and evaluates them using fairness, diversity, stability, safety, and robustness measures across policies, seeds, and pack settings.

3. Research Goal and Objectives

3.1 Research Goal

The goal of this thesis is to determine whether constrained procedural generation of new SAP-like artifacts can safely improve simulator-backed balance outcomes while preserving strategic diversity in a fixed, reproducible Super Auto Pets environment.

This goal is both methodological and empirical. Methodologically, the thesis aims to produce a reproducible balancing framework for a complex auto battler. Empirically, it aims to test whether generated content can improve balance-related outcomes without introducing unsafe or overly dominant artifacts.

3.2 Objectives

O1: Build a frozen, reproducible Super Auto Pets simulation environment suitable for controlled thesis evaluation.

O2: Design a constrained generated-artifact schema and deterministic injection mechanism.

O3: Define simulator-backed fairness, diversity, stability, safety, and efficiency metrics.

O4: Compare generated artifacts against no-generation and fixed-generation baselines.

O5: Compare one-shot and iterative balancing procedures.

O6: Evaluate robustness across policies, seeds, and packs.

Together, these objectives turn the main research goal into a clear sequence of tasks. First, the thesis builds a stable simulation environment in which balance can be studied reproducibly. It then defines how new artifacts can be generated and inserted into that environment in a controlled way. After that, it establishes the criteria used to judge whether those artifacts improve balance while remaining safe and preserving diversity. The remaining objectives focus on comparing the generated artifacts against meaningful baselines, testing one-shot and iterative generation strategies, and examining whether the results remain consistent across different policies, seeds, and pack settings.

4. Methodology

4.1 Frozen Simulator and Environment

This thesis uses a frozen SAP-compatible simulator rather than a live game client. That choice is both methodological and practical. A frozen simulator enables controlled repeated trials with fixed seeds, deterministic replay, explicit action masks, and episode-level metadata logging. Without that control, it would be difficult to separate a genuine balance effect from unrelated simulator drift or transient runtime behavior.

Rather than building the full Super Auto Pets rules engine from scratch, this thesis used a frozen vendored snapshot of the open-source `sapai` backend as the simulation core. This provided a stable SAP-compatible foundation while avoiding dependence on live patch drift or upstream package instability. The thesis contribution was therefore not the original game engine itself, but the deterministic wrapper built around that backend, together with generated-artifact injection, controlled policy interfaces, simulator-backed evaluation metrics, and experiment orchestration for reproducible thesis experiments.

The overall method combines system development with controlled empirical evaluation. The development component produced the environment wrapper, artifact-injection path, validators, experiment runners, and reporting tools. The empirical component then evaluated predefined conditions under repeatable procedures. This framing is consistent with software-engineering guidance that emphasizes explicit study design, execution procedure, and threats-to-validity reporting for reproducible empirical work (Runeson & Höst, 2009; Wohlin et al., 2012).

The environment wrapper exposes reset/step interaction, deterministic seeding, invalid-action accounting, and episode summaries. It is not intended to reproduce the entire live SAP ecosystem. Its role is to provide a stable experimental arena in which balancing hypotheses can be tested under controlled conditions.

4.2 Generated Artifact Schema and Injection

Generated artifacts were represented through a constrained schema that defines artifact type, tier, base statistics, trigger, target, effect, effect value, cost, and tags. Candidate artifacts then passed layered validation, including schema checks, hard feasibility checks, and safety-oriented checks. This kept the search space broad enough to affect play while excluding candidates that were clearly implausible or unsafe.

A key step in the project was moving from proxy-only artifact records to real artifact injection in the simulator. Valid artifacts were compiled into synthetic SAP pets and inserted into deterministic shop slots inside the frozen environment. As a result, generated artifacts were not only scored in the abstract; they became real content that agents could encounter and use during play.

4.3 Evaluation Protocol and Baselines

The empirical design compares generated artifacts against three baseline conditions: `no_generation`, `random_generation`, and `heuristic_generation`. These baselines are necessary because any claim of improvement is only meaningful if the comparison targets are valid, executable, and evaluated under the same protocol. This emphasis on explicit comparison rather than anecdotal success is consistent with prior automated balancing work (Beau & Bakkes, 2016; DeLaurentis et al., 2021).

Where appropriate, baseline and generator evaluations reuse paired pack, policy, and seed cells. This keeps comparisons aligned at the simulator-cell level rather than relying only on coarse aggregate means. The protocol records observed simulator metrics, comparison-to-reference metrics, and robustness summaries so that balance can be discussed in terms of fairness, diversity, stability, safety, and efficiency rather than only raw reward or win rate.

4.4 One-Shot, Iterative, and Transfer Procedures

Candidate artifacts were proposed through constrained sampling and mutation. Proxy metrics were still useful as cheap search guidance, but final thesis-facing selection was based on simulator results. The search loop distinguished between `best_record` and `best_safe_record`, which meant that a high scoring, but unsafe artifact could not become thesis evidence. This differs from the exact RL formulation used by Babin and Katchabaw (2025) and from the sequential RL framing used in PCGRL (Khalifa et al., 2020). That was a deliberate choice: this thesis prioritized explicit schema constraints, deterministic reproducibility, and transparent safety gating over end-to-end learned generation.

One-shot and iterative balancing were treated as genuinely different procedures. In the one-shot condition, a single artifact was generated and evaluated against the original reference state. In the iterative condition, an artifact was generated, injected into the updated content pool, and the search continued from that changed state. The iterative condition therefore represented a sequence of balancing interventions rather than just a different search mode.

Robustness and transfer were evaluated across policy families and pack conditions. The thesis-facing policies were random, heuristic, and `scripted_value`. The `scripted_value` policy is scripted rather than learned. This broadened the evaluation beyond a single simple baseline, but it also limited the strength of any claim about learned-opponent generalization.

4.5 Metrics and Safety Gates

The final evaluation combined fairness, diversity retention, and stability into a simulator-backed objective, while also tracking safety and efficiency. Fairness was based on comparison-to-reference behavior, diversity was measured from final team-signature entropy, and stability was

based on variability across matched policy and seed cells. Safety included the absence of backend errors and the rejection of degenerate patterns.

This design matters because the thesis is about balance as it appears in play, not just about the surface structure of generated artifacts. Unsafe exploratory winners can still appear during search, but the thesis-facing result must always be a safe accepted artifact. That distinction is central to the interpretation of Section 5. Appendix Section 10 gives the exact definitions and formulas for the reported terms.

5. Results

5.1 Primary Result: Balance Improvement Under Constraints

The primary research question asked whether constrained procedural generation could reduce imbalance while preserving diversity. The answer from the final matrix is a cautious yes.

The clearest comparison is the matched-protocol evaluation of the fixed heuristic baseline artifact against the generator's best safe artifact. Under that comparison, the heuristic baseline reached a final objective of 0.5294, while the generator's best safe artifact reached 0.5427. The generated artifact also produced slightly higher fairness (0.3581 versus 0.3479) and higher diversity retention (1.0116 versus 0.9646) while remaining safety compliant. The improvement is modest, not dramatic, but it is still a real improvement under the study's metric definitions.

A second supporting result came from the one-shot arm of the iterative study. That condition produced a safe artifact with a final objective of 0.5587 and a fairness score of 0.4168, which was stronger than the dedicated generator run's selected safe artifact. This suggests that the framework can discover safe artifacts with stronger balance-oriented performance than the specific generator winner reported in Table 1, although the most direct baseline comparison remains the heuristic-versus-best-safe result.

Table 1. Heuristic baseline versus generator best safe (matched protocol)

Candidate	Final objective	Fairness	Diversity retention	Stability	Mean outcome delta	Safety
Heuristic baseline	0.5294	0.3479	0.9646	0.6386	-0.0042	Pass
Generator best safe	0.5427	0.3581	1.0116	0.6395	0.0281	Pass

The main takeaway is that constrained procedural generation can improve the thesis balance objective while preserving high diversity, but the size of the improvement is limited. The correct claim is therefore feasibility and modest benefit, not full balance restoration.

5.2 One-Shot Versus Iterative Generation

The first secondary question asked whether iterative generation outperformed one-shot generation. The final matrix did not support that hypothesis.

The one-shot condition completed successfully and selected an artifact with a final objective of 0.5587, fairness 0.4168, diversity retention 0.9426, and stability 0.6007. The iterative condition found one safe artifact at step 1 with a final objective of 0.5531, fairness 0.3464, diversity retention 0.9840, and stability 0.7424, but it did not find a second safe artifact within the configured search budget and acceptance rules. Its final status was `no_safe_artifact_found`.

Table 2. One-shot versus iterative balancing

Condition	Status	Final objective	Fairness	Diversity retention	Stability	Safety
One-shot	ok	0.5587	0.4168	0.9426	0.6007	Pass
Iterative step 1	ok	0.5531	0.3464	0.9840	0.7424	Pass
Iterative final status	<code>no_safe_artifact_found</code>	-	-	-	-	-

This should be read as a real result rather than a failure of the infrastructure. The simulator and orchestration completed correctly, but the iterative procedure could not continue safely under the configured budget. Under this protocol, one-shot generation outperformed the tested iterative procedure.

Figure 1 brings the heuristic baseline, generator best safe result, one-shot winner, and iterative step 1 winner into a single comparison. The main pattern is clear: one-shot achieved the strongest fairness and final-objective values among the safe thesis-facing conditions, while the generator best safe and iterative step 1 results stayed closer to the heuristic baseline.

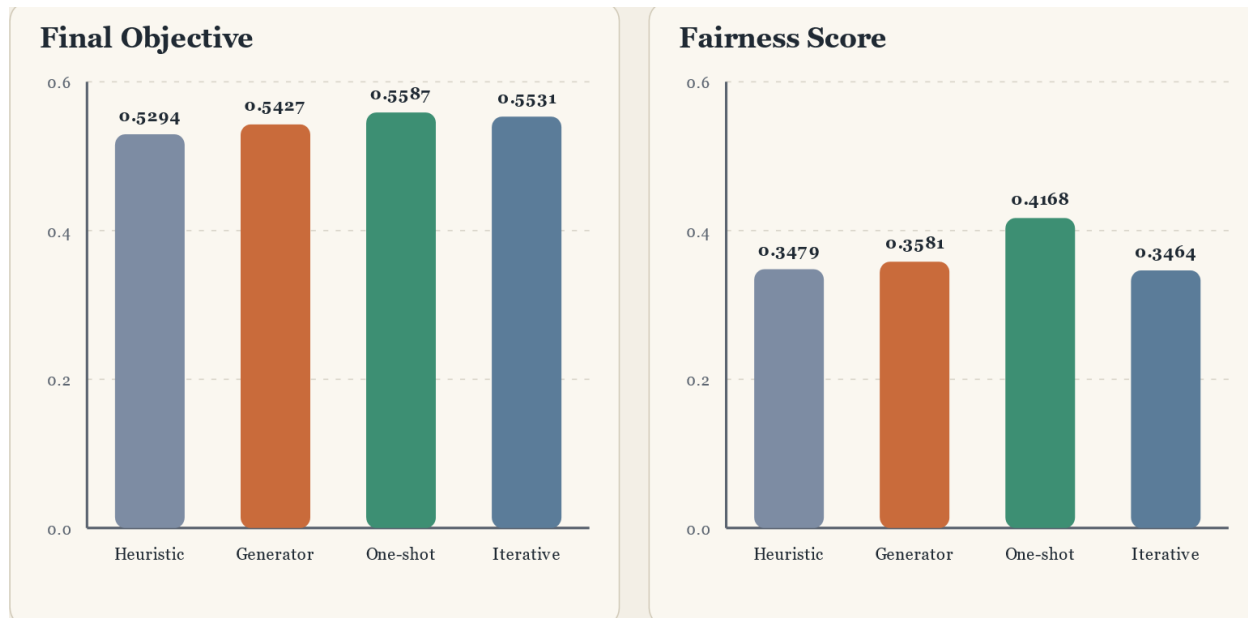


Figure 1. Final-matrix comparison of the safe thesis-facing conditions. The left panel compares final objective and the right panel compares fairness score for the heuristic baseline, generator best safe, one-shot selected, and iterative step 1 results.

5.3 Robustness and Transfer

The second secondary question asked whether selected artifacts generalized across opponent policies and random seeds. The results support a qualified robustness claim.

Posthoc cross-pack evaluation tested the selected artifacts on StandardPack and ExpansionPack1 across random, heuristic, and scripted_value policies with five seeds per pack. All three selected artifacts remained safety-compliant in these follow-up evaluations, and backend error rate remained 0.0 throughout the tested cross-pack runs. However, performance was not invariant.

Table 3. Cross-pack selected-artifact results

Artifact	Pack	Final objective	Fairness	Diversity retention	Mean outcome delta	Safety
Generator best safe	StandardPack	0.5465	0.3476	1.0090	0.0050	Pass
Generator best safe	ExpansionPack1	0.5236	0.3500	0.9701	0.0000	Pass
One-shot selected	StandardPack	0.5653	0.4406	0.9495	0.0133	Pass
One-shot selected	ExpansionPack1	0.5361	0.3859	0.9701	-0.0117	Pass
Iterative final set	StandardPack	0.5487	0.3452	1.0000	0.0100	Pass
Iterative final set	ExpansionPack1	0.5235	0.3444	0.9911	0.0117	Pass

The one-shot artifact showed the largest drop on ExpansionPack1, while the generator best safe artifact and the iterative final set were more stable across packs. The most accurate interpretation is that the selected artifacts transferred safely across the tested conditions, but their performance remained artifact-dependent rather than universal.

5.4 Additional Result: Safety Gates Matter

Another important result is that the highest-scoring artifact found during generator search was not thesis-eligible because it triggered backend-error behavior and failed the safety criteria. This validates the decision to distinguish `best_record` from `best_safe_record`. Without that distinction, the thesis could have reported a misleading success based on unsafe content.

This matters methodologically because it shows that even a constrained search space can still produce brittle or pathological artifacts when optimization pressure is high. For that reason, the safe artifact, not the unsafe exploratory winner, is the correct thesis-facing result.

5.5 Novelty of Results and Comparison to Related Work

The main novelty of these results is not that they show content generation always solves balance. The more important contribution is that they provide simulator-backed evidence that the content-addition framing from prior work can be carried into an auto battler under explicit safety constraints and reproducibility controls. Earlier work often treated balancing as an engineering process or as computational adjustment of existing systems (Beau & Bakkes, 2016; DeLaurentis et al., 2021). In contrast, this thesis evaluates the addition of new constrained content inside a deterministic SAP simulator.

The closest conceptual comparison is Babin and Katchabaw (2025), which argues that new content can be used to restore balance without directly editing older content. The present thesis differs methodologically because it uses constrained search with simulator-backed final acceptance in an auto battler rather than the exact generation setup used in that earlier work. The novelty here is therefore an adaptation result: the content-addition framing remained viable in SAP, but the observed improvement was modest, safety gating was essential, and iterative balancing was not automatically superior. That combination of a positive primary result and a negative iterative result is an important part of the contribution.

6. Discussion

6.1 Interpretation of the Main Findings

Taken together, the final matrix suggests that the content-addition framing from the supervisor's work does carry over to the auto battler setting, although only in a limited way. The central idea of adding constrained new content, injecting it into a simulator, and evaluating its effect through gameplay outcomes remained workable in SAP. This is the main positive contribution of the thesis.

At the same time, the results place clear limits on how strong the final claims can be. The gains were modest rather than transformative. One-shot generation performed better than the tested iterative procedure. Robustness across packs and policies was partial rather than universal. For that reason, the correct overall interpretation is cautious rather than sweeping.

6.2 Threats to Validity

Several threats to validity affect the interpretation of these results. First, the third policy family is `scripted_value`, not a trained learned opponent. This broadens the policy coverage beyond a single simple heuristic, but it does not support a claim of robustness against a truly learned adaptive policy. Second, the seed counts in the final matrix are lower than the ideal target considered earlier in the project plan, which limits the strength of the generalization claims. Third, the environment is a frozen snapshot of SAP-compatible mechanics, so the conclusions must be scoped to the controlled simulator rather than the live game ecosystem.

A further threat concerns metric construction. The thesis uses manually designed operational metrics for fairness, diversity retention, stability, and the final weighted objective. These definitions were developed for this study and refined during implementation so that the

comparisons would be transparent and reproducible, but they are not externally standardized measures of game balance. This means that some results may be sensitive to the chosen formulas, weights, and thresholds. That risk is partly reduced by reporting the component metrics alongside the combined objective and by basing the analysis on matched simulator outcomes rather than on proxy scores alone.

6.3 Limitations

The first major limitation is effect size. The improvements over the fixed heuristic baseline are real, but they are modest, so the thesis should be framed as evidence of feasible balance improvement rather than complete balance restoration. The second major limitation is that iterative balancing did not outperform one-shot generation under the configured budget and safety gate. This matters because it means the future-work direction suggested by the supervisor paper was not automatically confirmed in this domain. The third major limitation is that transfer was safe but not invariant, which shows that generated artifacts did not generalize equally well across all tested conditions.

A fourth limitation concerns statistical strength. The original project plan aimed for 20 or more seeds for the final key conditions if runtime allowed. In practice, the final matrix used eight evaluation seeds for the main generator and iterative runs and five seeds for transfer. This was a practical compromise between evidence strength and the cost of repeatedly evaluating simulator-backed conditions on local hardware, but it also weakens the strength of the generalization claims.

6.4 Generalizability and Implications

The results support a qualified form of generalizability. Within the tested simulator, selected artifacts transferred safely across multiple seeds, across three policy families, and across two pack settings. At the same time, effectiveness changed by artifact and condition. The strongest defensible conclusion is therefore that the approach generalizes partially within the tested simulator, not universally across all possible content conditions.

The main implication is that content addition can be a viable balancing mechanism in auto battlers when it is paired with reproducible simulator-backed evaluation and explicit safety constraints. A second implication is methodological: balance claims are much stronger when they are grounded in actual simulator outcomes rather than only in proxy objectives.

7. Conclusions

This thesis investigated whether constrained procedural generation of new SAP-like artifacts could improve simulator-backed balance outcomes while preserving strategic diversity in a fixed, reproducible Super Auto Pets environment. It addressed the stated objectives by building a deterministic SAP testbed (O1), implementing a constrained schema and injection mechanism (O2), defining simulator-backed metrics (O3), comparing against required baselines (O4), testing one-shot versus iterative balancing (O5), and evaluating robustness across policies, seeds, and packs (O6) within the broader balance-through-generation literature (Babin & Katchabaw, 2025; Beau & Bakkes, 2016; DeLaurentis et al., 2021).

The results provide a cautious positive answer to the primary research question. Safe generated artifacts modestly improved the thesis objective relative to the fixed heuristic baseline under a matched protocol (Section 5.1, Table 1, Figure 1). The results did not support the claim that iterative generation outperformed one-shot generation; one-shot was stronger, and the iterative procedure stopped at `no_safe_artifact_found` (Section 5.2, Table 2, Figure 1). The results supported only a qualified robustness claim: selected artifacts transferred safely across the tested policies, seeds, and packs, but effectiveness varied by artifact and condition (Section 5.3, Table 3). Section 5.4 also shows why the distinction between exploratory winners and accepted safe winners matters.

The thesis contribution is both methodological and empirical. Methodologically, it provides a reproducible simulator-backed framework for studying balance-through-generation in an auto battler. Empirically, it shows that the content-addition framing can be transferred to SAP in a limited but meaningful way. The strongest defensible conclusion is that constrained and safely evaluated generated content can produce modest balance improvement in the tested simulator, not that it solves SAP balance in general.

8. Future Work and Lessons Learnt

8.1 Future Work

Several future directions follow from the current results. The most immediate is to replace `scripted_value` with a genuinely learned opponent so that robustness can be tested against a stronger adaptive policy family. That would bring the question of robustness closer to the broader literature on strong game-playing agents in complex environments (Vinyals et al., 2019). Other useful next steps include increasing seed counts, expanding artifact support beyond the current subset, improving iterative search strategy and budget, and testing a broader range of packs and policy configurations.

Another useful extension would be to treat generated-artifact transfer as a first-class built-in experiment path rather than a post hoc analysis step. The current thesis now includes that evidence but integrating it more directly into the main orchestration would make the pipeline stronger.

8.2 Lessons Learnt

This thesis highlighted several lessons that may be useful for future work on automated game balancing.

1. Balance claims are much stronger when they are based on actual simulator results rather than only on estimated proxy scores. One of the most important developments in this project was moving from early proxy-based evaluation to real in-simulator testing of generated artifacts.
2. It is important to distinguish between artifacts that look promising during search and artifacts that are safe enough to report as results. Without that distinction, it is easy for the system to favor content that appears strong numerically but behaves poorly or unreliably in practice.

3. The iterative balancing idea turned out to be more difficult to apply effectively in an auto battler than initially expected. In this study, one-shot generation performed better because the iterative procedure was unable to find a second safe artifact within the configured search budget and safety constraints. This suggests that iterative balancing is not automatically the stronger approach in this genre of game.
4. Reproducibility was not just a technical convenience; it was essential to the credibility of the study. In a stochastic game setting, fixed seeds, controlled environment behavior, clear configuration tracking, and repeatable analysis are necessary if the final conclusions are to be trusted.

9. Acknowledgments

I would like to acknowledge Michael Katchabaw for his supervision throughout this thesis, and his preliminary work alongside Mathias on game balance through procedural content generation that this project expanded and used as a guiding foundation.

9.1 Use of Generative AI in Developing the Evaluation Criteria

I first developed an initial version of the evaluation criteria without using generative AI tools. This initial version identified the main goals of the study as fairness, diversity preservation, stability, safety, and reproducibility. The initial formulas, including their weights and threshold values, were chosen using my own design judgment based on the goals of the project, and were then refined iteratively through implementation and testing in the simulator. At that stage, however, some of the ideas and terms were still described only at a high level.

I then used generative AI tools to produce an alternative version of the evaluation criteria and their explanation. AI-assisted version was most useful in making the metric definitions more explicit, clarifying ambiguous terminology, and emphasizing the distinction between raw simulator outcomes, derived thesis metrics, and internal search-time values. It was also useful in helping me reflect on whether the manually chosen formulas and values were explained clearly and justified well enough for a reader.

Comparing the two versions showed that my original criteria had the right conceptual focus, but that the thesis needed clearer operational definitions and more precise language. In particular, it became clear that terms such as baseline, heuristic condition, fairness, diversity retention, and final objective needed to be defined explicitly so that the reader could understand exactly what was being measured and reported. This comparison also helped me recognize that manually chosen weights and thresholds needed to be presented as study-specific design choices rather than as universal balance measures.

What I learned from this process was that it was not enough to define the evaluation criteria only intuitively. I learned that the thesis needed to explain both why the metrics were chosen and how each reported value was calculated. The final evaluation criteria used in this thesis were therefore revised manually after this comparison.

10. Appendix

10.1 Key Terms Used in Results

The body of the thesis uses several technical terms that have specific meanings in this project. They are summarized here so that the reported values in Section 5 can be interpreted correctly.

- **Artifact:** The generated game content being evaluated. In the thesis-complete scope, this means a generated synthetic SAP-compatible pet with defined tier, base stats, trigger, target, effect, cost, and tags.
- **Policy:** The action-selection rule used to play the simulator. The thesis uses three policy families: random, heuristic, and scripted_value.
- **Random policy:** Selects a random valid action at each decision point.
- **Heuristic policy:** A simple deterministic greedy controller that buys the first affordable useful action in a fixed order, otherwise rolls, otherwise ends the turn.
- **Scripted-value policy:** A stronger state-aware scripted controller that scores candidate actions using team state, empty slots, combines, food value, and turn context. It is scripted, not learned.
- **Baseline condition:** A comparison condition used to judge whether generated content improves on a reasonable reference. The thesis uses no_generation, random_generation, and heuristic_generation.
- **no_generation:** No artifact is injected. This is the cleanest reference condition for measuring the effect of adding new content.
- **random_generation:** A randomly constructed valid artifact is injected and evaluated.
- **heuristic_generation:** A fixed hand-authored valid artifact is injected and evaluated. This is the strongest fixed content baseline in the thesis.
- **Reference condition:** The condition against which a candidate is compared when comparison metrics are computed. In the main thesis-facing comparisons, this is usually the matched no_generation run under the same pack, policy, and seed protocol.
- **One-shot generation:** A procedure that searches for and evaluates a single generated artifact against the original reference state.
- **Iterative balancing:** A procedure that generates an artifact, injects it into the updated content pool, and then searches again from that changed state.
- **Safe artifact:** A generated artifact that passes the final simulator-backed safety gate. Only safe artifacts are thesis-facing results.

10.2 Operational Definitions and Metric Formulas

The values reported in Section 5 are operationalized study metrics. They are not universal physical units or universal balance percentages. Their meaning comes from the exact definitions below.

In the formulas below, $\text{clip}(x, 0, 1)$ means values below 0 are set to 0 and values above 1 are set to 1.

Episode-level performance

For a single episode, the basic performance quantity is:

$$\text{outcome_score} = 0.75 * (\text{final_wins} / 10) + 0.25 * (\text{final_lives} / 10)$$

This gives more weight to wins than remaining lives, while keeping the score on a roughly 0 to 1 scale. It is a raw performance signal, not the final thesis objective.

Mean outcome score

For a condition evaluated over many episodes:

$$\text{mean_outcome_score} = \text{average}(\text{outcome_score over all episodes})$$

This is the average simulator performance for that condition.

Outcome-cell spread

The evaluation protocol groups episodes into matched cells defined by pack, policy, and comparison seed. For each cell, the mean outcome score is computed. `outcome_cell_spread` is then the mean absolute deviation of those cell-level means. Lower spread indicates more even behavior across evaluation cells.

Mean outcome delta

When a candidate condition is compared with a matched reference condition:

$$\text{mean_outcome_delta} = \text{candidate_mean_outcome_score} - \text{reference_mean_outcome_score}$$

Positive values mean the candidate performed better on average; negative values mean it performed worse.

Spread reduction and spread reduction percentage

$$\text{spread_reduction} = \text{reference_spread} - \text{candidate_spread}$$

$$\text{spread_reduction_pct} = \text{spread_reduction} / \max(\text{reference_spread}, 0.05)$$

Positive values mean the candidate reduced unevenness across cells. Negative values mean it worsened unevenness.

Performance retention

$$\text{performance_retention} = \text{clip}(1 - \text{abs}(\text{mean_outcome_delta}) / \max(\text{abs}(\text{reference_mean_outcome_score}), 0.10), 0, 1)$$

This rewards candidates that do not shift overall performance too violently relative to the reference. It does not say whether the shift is intrinsically good or bad; it only measures how large the shift is.

Fairness score

The thesis fairness score is defined as:

$\text{fairness_score} = \text{clip}(0.65 * \max(0, \text{spread_reduction_pct}) + 0.35 * \text{performance_retention}, 0, 1)$

This means fairness rewards two things:

1. reduced unevenness across matched cells, and
2. limited distortion of average performance relative to the reference.

The spread-improvement term is clipped at zero, so worsening spread does not produce a negative fairness contribution; it simply removes that reward.

Diversity retention

The thesis tracks diversity using entropy over final team signatures. A final team signature is a compact representation of the final team composition reached at the end of an episode.

$\text{diversity_retention} = \text{candidate_team_entropy} / \text{reference_team_entropy}$

Interpretation:

- 1.0 means diversity matched the reference.
- greater than 1.0 means the candidate produced slightly more diversity than the reference.
- less than 1.0 means the candidate reduced diversity.

Stability score

Robustness is summarized by comparing matched candidate-reference deltas across evaluation cells and policies. The thesis stability score is:

$\text{stability_score} = 1 / (1 + 4 * \text{cell_effect_std} + 2 * \text{policy_effect_std})$

Where:

- cell_effect_std is the standard deviation of the candidate-reference effect across matched evaluation cells.
- policy_effect_std is the standard deviation of policy-family mean effects.

Higher stability means the artifact's effect is more consistent across the tested conditions.

Final objective

The main thesis scalar summary is:

$\text{final_objective} = 0.60 * \text{fairness_score} + 0.20 * \text{clip}(\text{diversity_retention}, 0, 1) + 0.20 * \text{clip}(\text{stability_score}, 0, 1)$

This is the quantity reported in the main tables as the thesis balance objective.

The weighting reflects the priorities of the study:

- fairness is the primary goal, so it receives the largest weight;
- diversity retention matters because balance improvements should not collapse strategy variety;

- stability matters because a useful artifact should not only work in one narrow condition.

Safety gate

An artifact is considered thesis-facing safe only if it passes the final simulator-backed safety rules. Violations are flagged when any of the following occur:

- $\text{mean_outcome_delta} > 0.18$ -> dominance risk
- $\text{artifact_in_final_team_rate} \geq 0.80$ and $\text{mean_outcome_delta} > 0.08$ -> artifact saturation risk
- $\text{diversity_retention} < 0.75$ -> diversity collapse risk
- $\text{backend_error_rate} > 0.0$ -> simulator instability risk

Here, $\text{artifact_in_final_team_rate}$ is the fraction of evaluated episodes in which the injected artifact appears in the final team, and $\text{backend_error_rate}$ is the fraction of episodes that ended with a simulator exception or backend failure.

Only artifacts that pass this final safety gate are used as accepted thesis results.

10.3 Search-Time Values Versus Thesis-Facing Values

- **Proxy metrics:** Cheap estimates derived from the artifact specification itself. They help guide search but are not final evidence.
- **Proxy objective:** A weighted combination of proxy fairness, proxy diversity, and proxy stability used to guide sampling and mutation.
- **Selection objective:** The generator's internal ranking quantity, defined as:

$$\text{selection_objective} = \text{final_objective} + 0.1 * \text{proxy_objective} - \text{safety_penalty}$$

This is why a generator summary can show a larger best safe objective than the thesis table reports. The table reports final_objective , not the internal search ranking.

- **best_record:** The strongest candidate found by the search objective, even if it later fails the final safety gate.
- **best_safe_record:** The strongest candidate that also passes the final simulator-backed safety gate. This is the correct thesis-facing winner.

This distinction is important for interpreting Section 5.4. The thesis does not claim success based on the strongest unsafe artifact found during exploration. It claims success only from the strongest safe accepted artifact.

11. References

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